

COURSE CODE/TITLE:	CPE 212	SECTION:	CPE31S2
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ACTIVITY NO. 8.1 Install, Configure, and Manage Enterprise Availability Monitoring via Ansible

1. PROCEDURE OUTPUT

Create a playbook that installs Nagios in both Ubuntu and CentOS. Apply the concept of creating roles.

2. Describe how you did step 1. (Provide screenshots and explanations in your report. Make your report detailed such that it will look like a manual.)

For this lab activity, I decided to start by creating and testing individual playbook files for updating, downloading dependencies, and downloading the actual Nagios Core for both Ubuntu and CentOS.

First things first is setting the inventory file to be used. Mine is set up like so.

```
GNU nano 7.2 home_inventory.yaml
all:
  children:
    ubuntu_servers:
      hosts:
        ubuntu-node:
          ansible_host: 192.168.56.104
    centos_servers:
      hosts:
        centos-node:
          ansible_host: 192.168.56.103
```

Next is the playbook to update Ubuntu and CentOS just before installation.

```

GNU nano 7.2 update.yaml
--
- hosts: all
  become: true
  pre_tasks:
    - name: install updates (CentOS)
      tags: always
      dnf:
        update_only: yes
        update_cache: yes
      when: ansible_distribution == "CentOS"

    - name: install updates (Ubuntu)
      tags: always
      apt:
        upgrade: dist
        update_cache: yes
      when: ansible_distribution == "Ubuntu"

mdnlot@workstation:~/CPE212_loterte/hoa$ ansible-playbook -i home_inventory.yaml update.yaml --ask-become-pass
BECOME password:

PLAY [all] *****

TASK [Gathering Facts] *****
ok: [ubuntu-node]
ok: [centos-node]

TASK [install updates (CentOS)] *****
skipping: [ubuntu-node]
ok: [centos-node]

TASK [install updates (Ubuntu)] *****
skipping: [centos-node]
ok: [ubuntu-node]

PLAY RECAP *****
centos-node      : ok=2    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
ubuntu-node     : ok=2    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0

mdnlot@workstation:~/CPE212_loterte/hoa$

```

Next is the yaml file for downloading the dependencies required by Nagios Core for Ubuntu.

GNU nano 7.2

ubuntu_dependencies.yaml

```
---
- name: install Nagios dependencies on Ubuntu
  hosts: ubuntu_servers
  become: true

  tasks:
    - name: install
      ansible.builtin.apt:
        name:
          - build-essential
          - gcc
          - make
          - autoconf
          - automake
          - apache2
          - php
          - libapache2-mod-php
          - libgd-dev
          - libssl-dev
          - libperl-dev
          - libsnmp-perl
          - snmp
        state: present
        update_cache: true
      when: ansible_distribution == "Ubuntu"
```

[Read 25 lines]

```
ndm1ot@workstation:~/CPE212_loterte/hoa8$ ansible-playbook ubuntu_dependencies.y
aml -i home_inventory.yaml --ask-become-pass
BECOME password:

PLAY [install Nagios dependencies on Ubuntu] *****

TASK [Gathering Facts] *****
ok: [ubuntu-node]

TASK [install] *****
changed: [ubuntu-node]

PLAY RECAP *****
ubuntu-node : ok=2    changed=1    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0
```

Naturally, the next is installation of the Nagios dependencies for CentOS.



mdmlot@worksta

GNU nano 7.2

centos_

```
---
- name: Install Nagios Dependencies for CentOS
  hosts: centos_servers
  become: true

  tasks:
    - name: enable EPEL repository
      ansible.builtin.dnf:
        name: epel-release
        state: present

    - name: install
      ansible.builtin.dnf:
        name:
          - gcc
          - make
          - autoconf
          - automake
          - httpd
          - php
          - gd
          - gd-devel
          - openssl-devel
          - perl
          - perl-Net-SNMP
          - net-snmp
          - net-snmp-utils
        state: present

    - name: update package cache CentOS
      ansible.builtin.package:
        update_cache: true
      when: ansible_distribution == "CentOS"
```

```
mdm1ot@workstation:~/CPE212_loterte/hoa8$ ansible-playbook centos_dependencies.y
aml -i home_inventory.yaml --ask-become-pass
BECOME password:

PLAY [Install Nagios Dependencies for CentOS] *****

TASK [Gathering Facts] *****
ok: [centos-node]

TASK [enable EPEL repository] *****
changed: [centos-node]

TASK [install] *****
changed: [centos-node]

TASK [update package cache CentOS] *****
ok: [centos-node]

PLAY RECAP *****
centos-node      : ok=4    changed=2    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0
```

Next, is the actual download of Nagios Core for Ubuntu using the official asset download link from nagios.com

```
---
- name: install Nagios Core on Ubuntu
  hosts: ubuntu_servers
  become: true

  tasks:
    - name: Download and extract Nagios Core source
      ansible.builtin.unarchive:
        src: https://assets.nagios.com/downloads/nagioscore/releases/nagios-4.5.1.tar.gz
        dest: /tmp
        remote_src: true
        creates: /tmp/nagios-4.5.1
        when: ansible_distribution == "Ubuntu"
```

```
mdmlot@workstation:~/CPE212_loterte/hoa8$ ansible-playbook -i home_inventory.yaml ubuntu_download.yaml --ask-become-pass
BECOME password:

PLAY [install Nagios Core on Ubuntu] *****

TASK [Gathering Facts] *****
ok: [ubuntu-node]

TASK [Download and extract Nagios Core source] *****
changed: [ubuntu-node]

PLAY RECAP *****
ubuntu-node      : ok=2    changed=1    unreachable=0    failed=0    skipped=0
0      rescued=0    ignored=0
```

I ssh'd into the Ubuntu server to check if extraction is complete.

```
mdmlot@workstation:~/CPE212_loterte/hoa8$ ssh mdmlot@192.168.56.104
Welcome to Ubuntu 24.04.5 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

13 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

*** System restart required ***
You have mail.
Last login: Sat Sep 19 22:48:13 2026 from 192.168.56.105
mdmlot@server1:~$ ls -la /tmp/nagios-4.5.1/
total 640
drwxrwxr-x 21 root root 4096 Feb 28 2024 .
drwxrwxrwt 21 root root 4096 Sep 19 22:49 ..
-rwxrwxr-x 1 root root 346 Feb 28 2024 aclocal.m4
drwxrwxr-x 2 root root 4096 Feb 28 2024 autoconf-macros
drwxrwxr-x 2 root root 4096 Feb 28 2024 base
drwxrwxr-x 2 root root 4096 Feb 28 2024 cgi
-rw-rw-r-- 1 root root 35655 Feb 28 2024 Changelog
```

The next steps involve doing the same download and installation process on the CentOS server using the unarchive built in module.

```

---
- name: install Nagios Core on CentOS
  hosts: centos_servers
  become: true

  tasks:
    - name: Download and extract Nagios Core source
      ansible.builtin.unarchive:
        src: https://assets.nagios.com/downloads/nagioscore/releases/nagios-4.5.1.tar.gz
        dest: /tmp
        remote_src: true
        creates: /tmp/nagios-4.5.1
        when: ansible_distribution == "CentOS"

```

```

mdmlot@workstation:~/CPE212_loterte/hoa8$ nano centos_download.yaml
mdmlot@workstation:~/CPE212_loterte/hoa8$ ansible-playbook -i home_inventory.yaml centos_download.yaml --ask-become-pass
BECOME password:

PLAY [install Nagios Core on CentOS] *****

TASK [Gathering Facts] *****
ok: [centos-node]

TASK [Download and extract Nagios Core source] *****
changed: [centos-node]

PLAY RECAP *****
centos-node      : ok=2    changed=1    unreachable=0    failed=0    skipped=0
0      rescued=0    ignored=0

```

Using ssh also confirmed proper extraction of the Nagios Core files.

```

mdmlot@workstation:~/CPE212_loterte/hoa8$ ssh mdmlot@192.168.56.103
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Sat Sep 19 22:51:17 2026 from 192.168.56.105
[mdmlot@server3 ~]$ ls -la /tmp/nagios-4.5.1/
total 608
drwxrwxr-x. 21 root root  4096 Feb 28  2024 .
drwxrwxrwt. 20 root root  4096 Sep 19 22:52 ..
-rwxrwxr-x.  1 root root   346 Feb 28  2024 aclocal.m4
drwxrwxr-x.  2 root root  4096 Feb 28  2024 autoconf-macros
drwxrwxr-x.  2 root root  4096 Feb 28  2024 base
drwxrwxr-x.  2 root root  4096 Feb 28  2024 cgi
-rw-rw-r--.  1 root root 35655 Feb 28  2024 Changelog
drwxrwxr-x.  2 root root   148 Feb 28  2024 common
-rwxrwxr-x.  1 root root 43765 Feb 28  2024 config.guess
-rwxrwxr-x.  1 root root 36345 Feb 28  2024 config.sub
-rwxrwxr-x.  1 root root 259704 Feb 28  2024 configure
-rw-rw-r--.  1 root root 30493 Feb 28  2024 configure.ac

```

This next part involves the actual role implementation. I used ansible galaxy role to make the nagios_ubuntu and nagios_centos roles for both installation of dependencies and Nagios Core itself. The common role is used for updating both distros.

```
mdmlot@workstation:~/CPE212_loterte/hoa8$ ansible-galaxy role init nagios_ubuntu
- Role nagios_ubuntu was created successfully
mdmlot@workstation:~/CPE212_loterte/hoa8$ ansible-galaxy role init nagios_centos
- Role nagios_centos was created successfully
mdmlot@workstation:~/CPE212_loterte/hoa8$ ansible-galaxy role init common
^[[Bmdmlot@workstation:~/CPE212_loterte/hoa8$ ansible-galaxy role init common
- Role common was created successfully
mdmlot@workstation:~/CPE212_loterte/hoa8$ mkdir -p roles
mdmlot@workstation:~/CPE212_loterte/hoa8$ mv nagios_ubuntu nagios_centos common roles/
mdmlot@workstation:~/CPE212_loterte/hoa8$
```

This is the site.yml file that I used for Nagios Installation.

```
GNU nano 7.2                                     hoa8_site.yml *
```

```
---
- name: Apply updates to all managed nodes
  hosts: all
  become: true
  roles:
    - common

- name: Install and configure Nagios on Ubuntu servers
  hosts: ubuntu_servers
  become: true
  roles:
    - nagios_ubuntu

- name: Install and configure Nagios on CentOS servers
  hosts: centos_servers
  become: true
  roles:
    - nagios_centos
```

Through the main.yml files in the task directories per role, I transferred the actual plays, omitting the hosts:, become:, and when: declarations as they are now being handled in the site.yml. This one is the common tasks yml file.

```
GNU nano 7.2 roles/common/tasks/main.y
---
# tasks file for common
---
- name: Install updates (CentOS)
  dnf:
    update_only: yes
    state: latest
    update_cache: true
  when: ansible_distribution == "CentOS"
  tags: always

- name: Install updates (Ubuntu)
  apt:
    upgrade: dist
    update_cache: yes
  when: ansible_distribution == "Ubuntu"
  tags: always
```

This one is for the nagios_ubuntu role.

```
GNU nano 7.2 main.yml *
---
# tasks file for nagios_ubuntu
- name: Install Nagios dependencies on Ubuntu
  ansible.builtin.apt:
    name:
      - build-essential
      - gcc
      - make
      - autoconf
      - automake
      - apache2
      - php
      - libapache2-mod-php
      - libgd-dev
      - libssl-dev
      - libperl-dev
      - libsnmp-perl
      - snmp
    state: present
    update_cache: true

- name: Download and extract Nagios Core source
  ansible.builtin.unarchive:
    src: https://assets.nagios.com/downloads/nagioscore/releases/nagios-4.5.1.tar.gz
    dest: /tmp
    remote_src: true
    creates: /tmp/nagios-4.5.1
```

Next is the nagios_centos role's task main.yml.

```
GNU nano 7.2                               ./roles/nagios_centos/tasks/main.yml *
---
#tasks file for nagios_centos
- name: Enable EPEL repository
  ansible.builtin.dnf:
    name: epel-release
    state: present

- name: Install Nagios dependencies on CentOS
  ansible.builtin.dnf:
    name:
      - gcc
      - make
      - autoconf
      - automake
      - httpd
      - php
      - gd
      - gd-devel
      - openssl-devel
      - perl
      - perl-Net-SNMP
      - net-snmp
      - net-snmp-utils
    state: present

- name: Download and extract Nagios Core source
  ansible.builtin.unarchive:
    src: https://assets.nagios.com/downloads/nagioscore/releases/nagios-4.5.1.tar.gz
    dest: /tmp
    remote_src: true
    creates: /tmp/nagios-4.5.1
```

3. Show an output of the installed Nagios for both Ubuntu and CentOS.

```

skipping: [ubuntu-node]
ok: [centos-node]

TASK [common : Install updates (Ubuntu)] *****
skipping: [centos-node]
ok: [ubuntu-node]

PLAY [Install and configure Nagios on Ubuntu servers] *****

TASK [Gathering Facts] *****
ok: [ubuntu-node]

TASK [nagios_ubuntu : Install Nagios dependencies on Ubuntu] *****
ok: [ubuntu-node]

TASK [nagios_ubuntu : Download and extract Nagios Core source] *****
skipping: [ubuntu-node]

PLAY [Install and configure Nagios on CentOS servers] *****

TASK [Gathering Facts] *****
ok: [centos-node]

TASK [nagios_centos : Enable EPEL repository] *****
ok: [centos-node]

TASK [nagios_centos : Install Nagios dependencies on CentOS] *****
ok: [centos-node]

TASK [nagios_centos : Download and extract Nagios Core source] *****
skipping: [centos-node]

PLAY RECAP *****
centos-node      : ok=5    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
ubuntu-node     : ok=4    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0

mdmlot@workstation:~/CPE212_loterte/hoa8$

```

```

mdmlot@server1:~$ ls /tmp/nagios*
aclocal.m4      debian          LICENSE         t
autoconf-macros docs            Makefile.in    tap
base            doxy.conf      make-tarball   test
cgi             functions      mkpackage      THANKS
Changelog       html           module          t-tap
common          include        nagios.spec    update-version
config.guess    indent-all.sh nagios.sysconfig UPGRADING
config.sub      indent.sh      pkginfo.in     worker
configure       INSTALLING    README.md      xdata
configure.ac    install-sh    sample-config
contrib         LEGAL         startup
CONTRIBUTING.md lib            subst.in
mdmlot@server1:~$

```

```
systemd-private-20b8d05d7ca9481c8724c0c0005a209a-upower.service-C7Y1Lg
mdmlot@server3 ~]$ ls /tmp/nagios-4.5.1
local.m4      debian      LICENSE     t
autoconf-macros docs        Makefile.in tap
base          doxy.conf  make-tarball test
gi            functions  mkpackage   THANKS
hangelog      html       module      t-tap
common        include    nagios.spec update-version
onfig.guess   indent-all.sh nagios.sysconfig UPGRADING
onfig.sub     indent.sh  pkginfo.in  worker
onfigure      INSTALLING README.md    xdata
onfigure.ac   install-sh sample-config
ontrib        LEGAL      startup
CONTRIBUTING.md lib         subst.in
mdmlot@server3 ~]$
```

4. Make sure to create a new repository in GitHub for this activity.

https://github.com/qmdmlot/CPE212_loterte/tree/main/hoa8

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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